Multitoken Hallucination Probes

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Contents

1 Motivation **1**

2 Research Proposal **1**

3 Probing Performance With Temporal Context **2**

4 Attention Probes **3**

5 Paper reproduction **4**
5.1 Llama-3.1-8b-Instruct 4
5.2 Apertus-8B-2509-Instruct 5

6 Contributions **6**

7 Summary **7**

1 Motivation

Recent work by Obeso et al [5] introduces a streaming, token-level hallucination detection framework that attaches lightweight linear probes to intermediate hidden states of large language models (LLMs). Their key finding is that hidden states contain rich information about whether a token belongs to a factually supported or hallucinated entity. Linear probes trained on entity-level labels substantially outperform uncertainty-based baselines in long-form generation, establishing internal activation monitoring as a viable alternative to expensive external-verification pipelines [5].

A possible limitation in this work is that a single hidden state is not always a sufficient summary of the relevant context. This is especially true for multi-token entities, where no single token cleanly encodes the entire structure (e.g. "New York City"). Additionally, hallucinated continuations often allocate attention inward (reinforcing previously generated hallucinated content), whereas correct answers place relatively more attention weight on question tokens and other grounding context [3]. These insights suggest that hallucination detection may benefit from explicitly incorporating a small set of past context tokens rather than relying on a single token's hidden state [1], as was also shown by McKenzie et al [3] in their study of High-Stakes Interactions.

2 Research Proposal

Motivated by these observations, we propose an extension to the token-level detection framework that conditions probes on the model's own attention over prior tokens. Specifically, at each generation step we:

[1] Here, a single hidden state refers to the model's internal representation at one token position (typically the current or final token), rather than a set or sequence of hidden states from multiple past tokens.

1. Extract the attention distribution over previous tokens.
2. Select the top-k most attended tokens (i.e., those most influential in producing the current token).
3. Concatenate their hidden states to form an expanded contextual representation.
4. Feed this representation into a linear probe that predicts whether the current token belongs to a supported or hallucinated entity.

By conditioning probes on these high-attention token embeddings, we aim to capture patterns of grounding vs. self-reinforcement [2] that single-state probes cannot. Our hypothesis is that attention-informed probing will improve detection particularly for:

- multi-token entities whose meaning is distributed across several tokens;
- hallucinations that emerge gradually through self-reinforcement.

In addition to the top-k attention-based context selection described above, we also evaluate attention probes [2] (EleutherAI attention probes blog) as an alternative probing strategy. Rather than explicitly selecting or concatenating tokens, attention probes use an attention layer to combine prior hidden states when producing a token-level prediction. We evaluate these probes in the same per-token, streaming setting as the base framework, enabling direct comparison with single-state and pooling-based probes.

This research direction directly builds on and extends the findings of Obeso et al. (2025), and McKenzie et al [3], aiming to enrich probe inputs with a small, model-selected contextual window rather than relying solely on the final hidden state. We also rely on the following attention probes implementation: (EleutherAI attention probes blog).

3 Probing Performance With Temporal Context

In these experiments, we trained a series of token-level hallucination probes on the same underlying Llama-3.1-8B model while varying only the amount of activation history provided to the probe. Specifically, we compared a linear probe with no temporal context to probes that concatenate hidden states from the last 2, 5, and 10 forward passes of a target layer. The overall picture is that the improvements from adding more context are present but modest. Across all four datasets, the 2-, 5-, and 10-step context variants form a clear ordering, but the deltas are small rather than transformative. The linear probe baseline remains surprisingly strong, and even the larger-context models only pull ahead by a narrow margin.

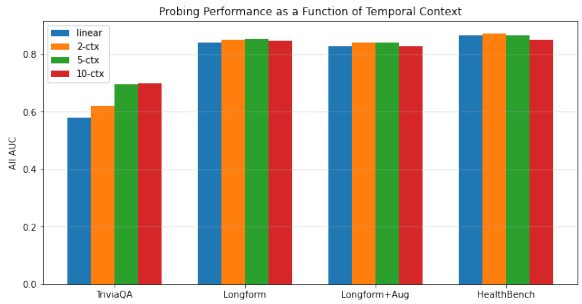

Figure 1: AUC across context windows.

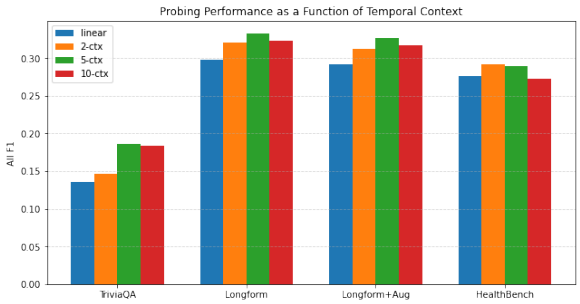

Figure 2: F1 across context windows.

Both F1 and AUC show the same trend: a consistent but shallow slope rather than a step-change. This suggests that while additional history helps the probe make slightly more stable decisions, the model is already capturing

[2] Here, self-reinforcement refers to the model allocating disproportionate attention to its own previously generated tokens, thereby reinforcing an internally consistent but potentially incorrect continuation instead of grounding the prediction in the input question or external context.

most of the usable signal from a very short window. In other words, the gains are real but not large enough to indicate a fundamentally different regime of performance.

One likely reason for the limited gains is that the hidden states at a single layer already encode a highly compressed summary of the recent context, so explicitly concatenating more steps does not introduce much genuinely new information. Additionally, the probe architecture itself is relatively simple, meaning it may not fully exploit long-range temporal structure even when provided.

4 Attention Probes

Method overview. Attention probes [2] provide an alternative to fixed pooling or token selection by learning how to aggregate information from prior hidden states. Given the hidden states of previously generated tokens, an attention probe applies a small set of learned query parameters to compute attention weights over the past context. Each attention head assigns a single weight to every prior token, and these weights are used to form a weighted sum of value projections of the hidden states. The resulting aggregation directly produces the probe output, without a separate classifier or pooling stage. In our setting, attention probes are applied in a per-token, streaming manner: at each generation step, the probe attends only to previously generated tokens and predicts whether the current token belongs to a supported or hallucinated entity. This formulation allows the probe to flexibly emphasize context tokens that are most informative for hallucination detection, while remaining lightweight and directly comparable to standard token-level linear probes. Our implementation is additionally informed by the attention probe formulation described in a technical EleutherAI attention probes blog.

Formally, for each token position t and head k, attention logits over prior tokens are computed as

$$\ell_{t,i}^{(k)} = (W_q^{(:,k)})^\top \mathbf{h}_i + b_k \cdot \max(t-i, 0),$$

where b_k is a learned scalar positional weight that induces a causal bias toward earlier tokens. Attention weights are obtained via a softmax over the causal context,

$$\alpha_{t,i}^{(k)} = \frac{\exp(\ell_{t,i}^{(k)})}{\sum_{j \le t} \exp(\ell_{t,j}^{(k)})}.$$

In parallel, value vectors are computed as

$$\mathbf{v}_i^{(k)} = W_v^{(k)} \mathbf{h}_i,$$

and aggregated to produce per-head probe outputs,

$$\mathbf{z}_t^{(k)} = \sum_{i \le t} \alpha_{t,i}^{(k)} \mathbf{v}_i^{(k)}.$$

The final probe output for token t is obtained by averaging over heads,

$$\mathbf{z}_t = \frac{1}{K} \sum_{k=1}^{K} \mathbf{z}_t^{(k)},$$

yielding a scalar prediction for each token position. The probe operates in a streaming manner, attending only to previously generated tokens, and directly produces per-token predictions without a separate classifier or pooling stage.

Experimental evaluation. We evaluated the attention probes in a comparative setup against standard linear probes across four datasets: LongFact, LongFact-augmented, TriviaQA, and HealthBench. Each probe was trained to detect hallucinated tokens in a per-token streaming manner on the Meta-Llama-3.1-8B-Instruct model, using an identical underlying language model, tokenizer, and training configuration for both probe types. For all experiments, we report both span-level F1 scores (`span_f1`) and area under the precision–recall curve

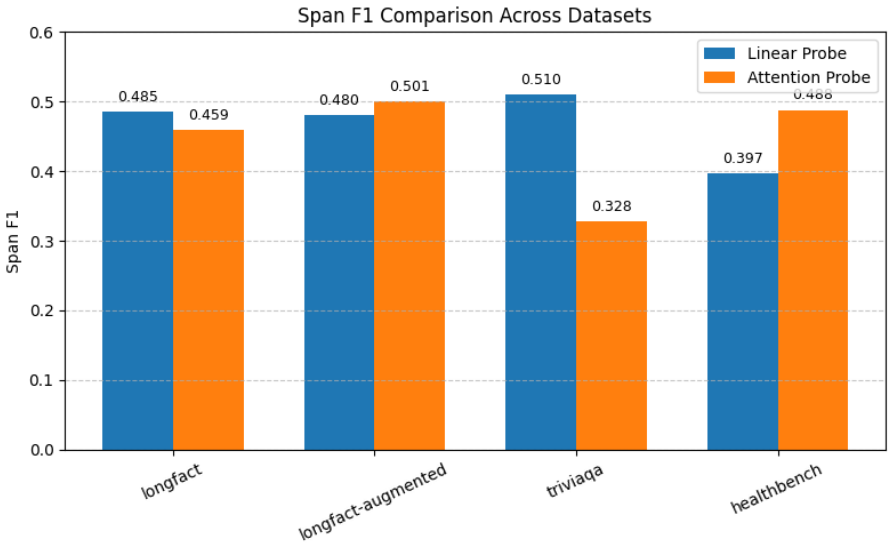

Figure 3: Span F1 comparison of Linear and Attention probes across datasets. The bars show side-by-side performance, with numeric values annotated above each bar for clarity. Attention probes outperform linear probes on LongFact-augmented and HealthBench, while linear probes perform better on TriviaQA.

(`span_auc`) to capture probe performance in terms of exact span matching and overall token-level discrimination, respectively.

The `span_f1` metric measures the F1 score at the level of predicted spans, treating each contiguous sequence of predicted hallucinated tokens as a single unit. The `span_auc` metric measures how well the probe ranks hallucinated spans above supported ones across all possible decision thresholds, complementing F1 metrics that rely on a fixed classification threshold.

Figures 3 and 4 illustrate the comparative performance of linear and attention probes across datasets. In Figure 3, attention probes slightly outperform linear probes on LongFact-augmented (0.501 vs. 0.480) and HealthBench (0.488 vs. 0.397), while linear probes achieve higher span_f1 on TriviaQA (0.510 vs. 0.328). Figure 4 shows that linear probes achieve higher span_auc on most datasets except TriviaQA, where the attention probe substantially exceeds linear performance (0.915 vs. 0.851). The plots clearly visualize these trends, with side-by-side bars for each probe type and numeric values annotated for precise comparison.
Overall, the results indicate that linear probes constitute a strong baseline, achieving consistently high performance across datasets. Attention probes are competitive with, and in some cases surpass, linear probes, particularly on augmented or more context-dependent datasets, demonstrating their ability to selectively aggregate contextual information to improve hallucination detection. The combined view of `span_f1` and `span_auc` suggests that attention probes can yield improved span-level detection in some settings where flexible context aggregation is important, while linear probes remain a strong and robust baseline, often exhibiting superior overall ranking performance on more straightforward datasets.

5 Paper reproduction

For additional evaluation of the methods proposed in Obeso et al. [5], we decided that we will reproduce the findings in the paper and try to adapt the described method on an open-source Apertus LLM [1] to allow for easier work on hallucinations in the future.

5.1 Llama-3.1-8b-Instruct

In the initial part of the paper reproduction we tried to reproduce results for Llama-3.1-8b-Instruct from Meta [4]. We tried to do so, to ensure that our improvements of the methods proposed in the paper are compared to

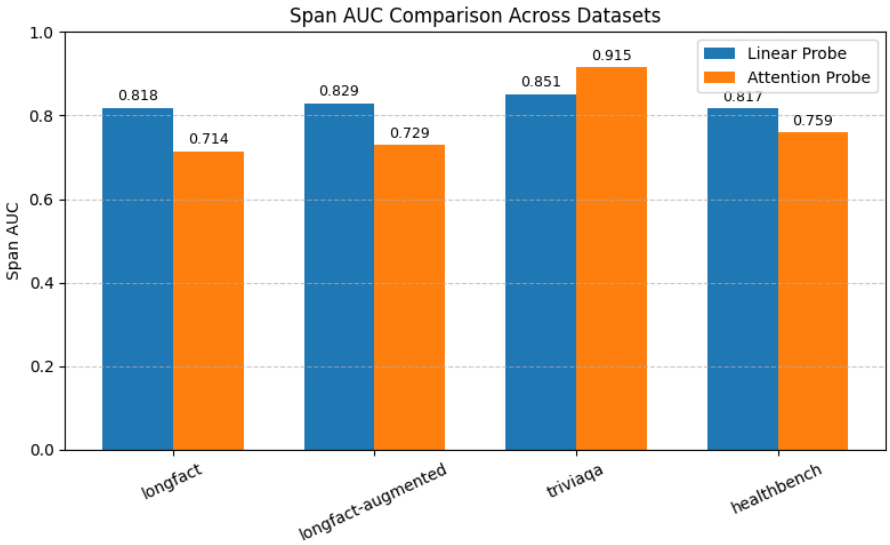

Figure 4: Span AUC comparison of Linear and Attention probes across datasets. Attention probes achieve higher span AUC on TriviaQA, while linear probes dominate on other datasets. The numeric annotations above each bar indicate exact values for precise comparison.

the correct baselines. Using the same datasets and by matching hyperparameters from the paper, we managed to achieve the similar result when it comes to quality of hallucinations detection.

Table 1: AUC Comparison for probes trained on their dataset: Linear Probe (No LoRA) vs. LoRA Probe on Llama Models. The performance of our probes is very similar, expect for the TriviaQA dataset. We have used the same approach to this dataset, even though it is short form and we suspect this might have reduced the achieved performance.

	Reference (Llama-3.1-8B)		**Ours (Llama-3.1-8b)**	
Dataset	**Linear**	**LoRA**	**Linear**	**LoRA**
LongFact	0.8535	0.8938	0.8280	0.8945
HealthBench	0.8560	0.8960	0.8645	0.9296
TriviaQA	0.9179	0.9651	0.5780	0.7131

5.2 Apertus-8B-2509-Instruct

Reproduction of the results for Apertus-8B-2509-Instruct was much more difficult, because we had to prepare the model's generations with annotated hallucination spans by ourselves. It turned out that this process is not straightforward. The original codebase of the paper does not contain any script for generating cohesive model outputs with answers for given questions. What is more, there was no direct mapping between final annotated hallucinations with input prompts.

Additionally, we had no access to Anthropic API, which was used to annotate the hallucination spans, so we had to adapt the scripts for OpenAI API, to which we had access.

To overcome these issues we generated model's responses using transformers library. To achieve this we used two prompt datasets - LongFact, consisting of over 2k prompts from different topics [6] and LongFact Augmented, which is a larger version of LongFact, developed by Obeso et al. Then with OpenAI API with websearch feature, we annotated the generations achieving 22k datapoints in total for each of the models.

Manual comparison of few randomly sampled annotated outputs from the original dataset and from our annotations suggested that our pipeline was correct.

Initial training of the probe, using the LongFact Augmented for training, and testing on LongFact dataset failed. The probe trained in this way collapsed to negative-only responses for the test set. To verify if this problem lay with our pipeline, we split the LongFact Augmented dataset into training and test sets. It turned out that the probes trained on Llama were actually able to learn to detect hallucinations with this setup, hinting that LongFact and LongFact Augmented might come from different distributions. In Fig. 5, we report the AUC scores comparing the performance of probes trained on Apertus and Llama. To compare the probes, we trained both on the dataset generated by us, using our annotation pipeline. It turns out that the probe trained on Apertus performs worse in overall. Unfortunately, we weren't able to identify a reason for this behavior. Our initial guess is that the Apertus responses differ significantly from Llama's, and our reproduction pipeline failed in this setup. Another reason for such behavior could be that the internal representations of Apertus are less verbose or simply hallucinate less—we haven't evaluated the datasets quantitatively.

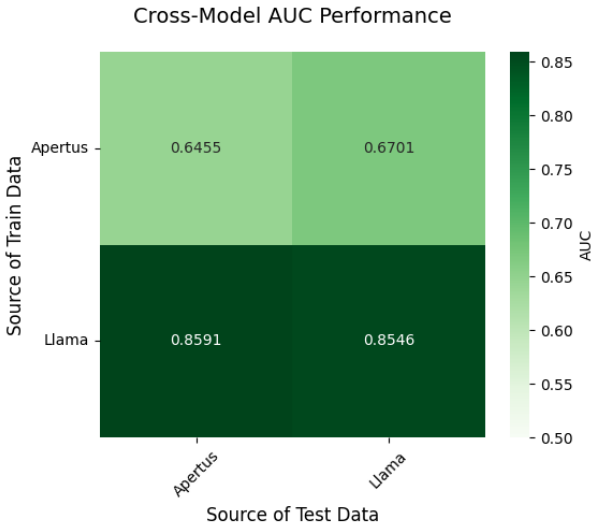

Figure 5: AUC for cross-model performance between Apertus-8B-2509-Instruct and Llama-3.1-8b-Instruct. Both probes were trained without LoRA adapters and on the same split of longfact augumented. Performance of probe trained on Apertus is significantly worse.

6 Contributions

Tymotuesz Kwiecinski prepared the environment on the Clariden cluster and worked on reproducing the paper for Llama-3.1-8b-Instruct and Apertus-8B-2509-Instruct. He also analysed the results of the reproduction and prepared the corresponding sections in the report.

Michal Korniak implemented the attention probes, conducted the related experiments, and performed the comparative evaluation against linear probes across datasets. He also analyzed the results and authored the corresponding sections of the report describing the experimental setup and findings.

Klejdi Sevdari designed and conducted the temporal context experiments, evaluating the impact of multi-step activation history on token-level hallucination detection. He set up and maintained the experimental repository and also contributed to the analysis and interpretation of results and to the overall structure of the report.

7 Summary

We have successfully reproduced the paper and proposed new improvements of the attention probes. Even though the modifications that we proposed do not show improvements we anticipated, but still are a good research direction, which should be explored.

References

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